



Design Report – IFS09 Telemetry Receiver

Department: TE

Author: Andrés Sánchez de Ágreda

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1. Executive Summary & Functional Description

This Design Report responds directly to the requirements set forth in the IFS09 Telemetry Receiver Work Order (WO_TE-IFS09-TEL-001). The telemetry receiver serves as the primary trackside base station interface for the ISC Formula Student race car during competition testing and dynamic events.

The board captures high-frequency 2.4 GHz wireless telemetry packets (CAN-bus diagnostics, powertrain parameters, battery management data, and vehicle dynamics) broadcast by the car, verify packet integrity, and streams the uncorrupted data over a high-speed USB-C serial interface (115200 baud) to the pit wall workstation.

2. System Architecture Diagram

2.1. Why a Separate nRF24L01+ Module with PA + LNA and SMA Connector Was Implemented

During the Formula Student Austria (FSA 2026) and Formula Student Spain (FSS 2026) competitions, the previous telemetry architecture suffered critical range limitations, frequent packet dropouts, and blind spots around large track perimeters. These issues stemmed from:

1. Low transmit power and poor receiver sensitivity of standard integrated PCB-trace antennas.

Solution in IFS09:

- A dedicated nRF24L01+ module with integrated Power Amplifier (PA) and Low-Noise Amplifier (LNA) was chosen over standard PCB-antenna modules.
- Link Margin Improvement: The integrated LNA dramatically increases receiver sensitivity down to -95dBm (@ 250kbps), while the PA provides up to $+20\text{dBm}$ output power, eliminating past dropouts.
- External SMA Connector (50Ω): Enables connection of a high-gain, omnidirectional 2.4 GHz whip antenna elevated outside of the pit box enclosure, ensuring line-of-sight propagation over distances $\geq 300\text{m}$.
- RF Decoupling: Tightly coupled bypass capacitors ($1\mu\text{F}$ C1 and 10nF C2) were placed immediately adjacent to the transceiver header pins (U1) to filter high-frequency switching noise.

2.2. Why a Dedicated External 3.3V Regulator (LM1117MP-3.3) Was Used

The Arduino Nano 3.0 onboard 3.3V rail is supplied internally by its USB-to-UART bridge (or a small auxiliary regulator), which has a maximum current limit of approximately 50mA.

- Current Demand Analysis: The nRF24L01+ PA+LNA transceiver draws significant burst currents during active reception and amplification pulses—ranging between 115mA and 150mA.
- Problem Avoided: Attempting to power the PA+LNA module directly from the Arduino Nano's 3.3V pin causes immediate rail sag, voltage brownouts, SPI communication freezes, and LNA stage failure.
- Circuit Implementation: A dedicated LM1117MP-3.3 Low-Dropout (LDO) regulator (U2, SOT-223 package) was implemented. It steps down the 5V USB bus to an isolated, high-current 3.3V plane (rated up to 800mA).
- Capacitor Network: Per datasheet specifications, a $10\mu\text{F}$ input electrolytic/ceramic capacitor (C5) and a $10\mu\text{F}$ output capacitor (C4) ensure transient stability and low output ripple ($< 30\text{mV}_{p-p}$).



3. Hardware & Design Requirements

- Microcontroller Module: Standard socket/footprint to mount an Arduino Nano 3.0 featuring a native USB-C connector for communication and 5V bus power.
- RF Transceiver Interface: Standard 2x4 (2.54 mm pitch) socket header to mount the nRF24L01+ PA+LNA transceiver module.
- External Antenna Connection: Edge-mounted / chassis-mounted SMA Female (Jack) connector (50 Ω impedance) matched to a high-gain 2.4 GHz omnidirectional whip antenna.

3.1. Pin Mapping & Interface Specification

Processing Unit (A1): Arduino Nano 3.0 with native USB-C connector (ATmega328P @ 16 MHz, 5V logic). A 100nF capacitor (C3) was added from AREF to GND to stabilize analog references.

RF-NANO Pinout Standard Alignment: To prevent firmware mismatches across team development boards, the SPI and control lines follow the standard RF-NANO pin mapping:

- CE (Chip Enable): Connected to Arduino D7 (PD7).
- CSN (Chip Select Not): Connected to Arduino D8 (PB0).
- Hardware SPI: MOSI $\rightarrow$ D11, MISO $\rightarrow$ D12, SCK $\rightarrow$ D13.

3.2. Testability & Mechanical Requirements

To simplify trackside troubleshooting and isolate software bugs from hardware connection issues:

1. Comm_status_indicator (LED D1 + 2k Ω resistor R1): Connected to digital pin D2. Configured via software to blink during active packet reception or stay solid when a telemetry link is active.
2. nRF24_detected (LED D2 + 2k Ω resistor R2): Connected to digital pin D3. Driven HIGH upon successful verification of radio.isChipConnected() in firmware setup().
3. Power Test Point (TP1 / nrf_power_tp): Dedicated top-layer test pad located directly on the nRF24 3.3V supply rail for multimeter/oscilloscope probing.

4. Verification & Testing Protocol

The PCB layout, DRC/ERC checks, and schematic export (Schematics-2.0.pdf) are 100% complete. Because the PCB has not yet been manufactured/ordered, physical bench measurements and track testing will be executed upon arrival of the assembled prototypes.

Every manufactured board unit must successfully pass the following verification tests in sequence:

4.1. Power Supply Rails Verification

Procedure: Before powerup check continuity in the supply circuit for both the Arduino & the nrf24 module, then power up and check led lights.

4.2. Microcontroller to Transceiver Interface Test

Procedure: Flash SPI test firmware. Execute radio.begin() and check radio.isChipConnected() over serial monitor.

Acceptance Criteria: Transceiver returns TRUE status, valid register dumps, and clean clock waveforms on TP_SCK.

4.3. USB-Serial Communication Test with PuTTY

Procedure: Connect the board to PC via USB-C. Open PuTTY terminal at 115200 baud, 8-N-1, No Flow Control.

Acceptance Criteria: Continuous streaming of raw telemetry frames for 10 minutes with 0% framing errors or data corruption.

4.4. Vehicle Dynamic RF Range & Link Test



Procedure: Install telemetry transmitter in the car or dynamic test fixture; place receiver in the pits with external 2.4 GHz antenna.

Acceptance Criteria: Packet Reception Rate (PRR) $\geq 95\%$ at line-of-sight distance ≥ 300 meters under race-condition electrical noise